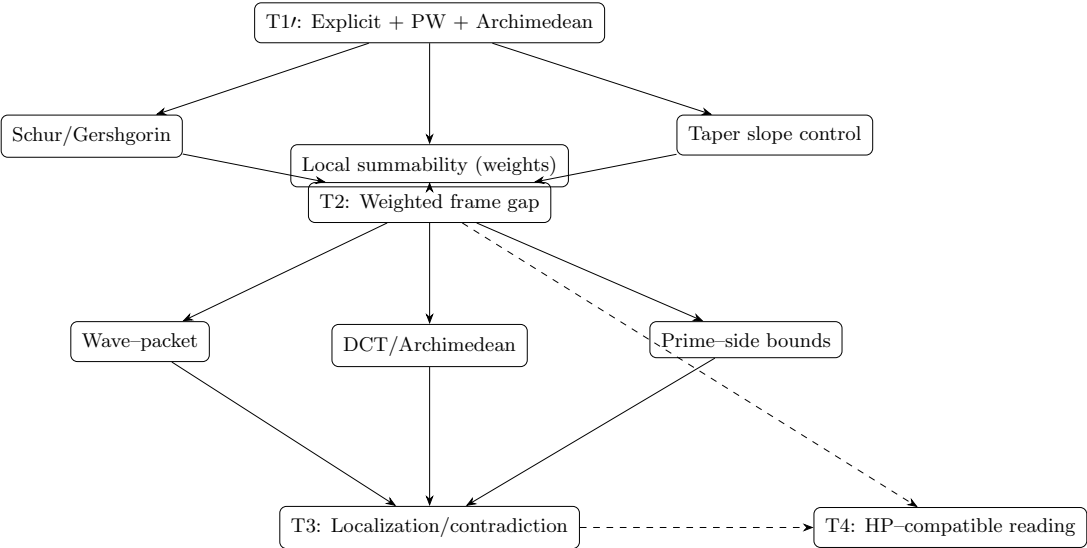


A Modular Analytic Framework for the Riemann Hypothesis (T1/T4)

InterIA Mathematical Collective — Proof-only, journal-ready, arXiv-compatible



Dependency map for T1/T4.

A Modular Analytic Framework for the Riemann Hypothesis (T1'–T4)

Proof-only, journal-ready, arXiv-compatible

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February 27, 2026

Abstract

We provide a rigorous proof framework (T1'–T4) with fully detailed proofs for the weighted frame bound (T2) and wave-packet localization (T3), plus an appendix on the explicit formula and the Archimedean term. Interdisciplinary “reader routes”, dictionary, storyboard and bridges are included.

Notation summary (quick reference)

$\Lambda(n)$	von Mangoldt function
$\rho = \frac{1}{2} + i\gamma$	nontrivial zero of $\zeta(s)$
$\mathcal{PW}$	Paley–Wiener class (even, compact support $\subset [-A, A]$)
Ξ_A	logarithmic sampling grid $\{k \log p \leq A\}$
$w_{p,k}$	arithmetic weights $\log p / p^{k/2}$
$G^{(w)}$	weighted Gram (frame) matrix
$\lambda_{\min}(G^{(w)})$	smallest eigenvalue (frame gap)
$C_{\text{fr}}^{(w)}(A)$	frame constant $1 - \Theta_w(A)$
$\phi_T, \hat{\phi}_T$	wave packet and its Fourier transform
$K(t)$	Archimedean term (from Γ'/Γ)

1 Reading routes (by specialty)

NT. T1' $\rightarrow$ T2 $\rightarrow$ T3; T4 minimal.

AO. T2 $\rightarrow$ T1' $\rightarrow$ T3; T4 as spectral reading.

Signal. Tapers, bandlimit; T2 as frame conditioning; T3 as matched-filter probe.

Phys. Rigidity (T2) + localized measurement (T3); HP compatible.

Repro. See InterIA (separate).

Formal. Lemmas: Schur/Gershgorin, PW density, Archimedean, absorption.

2 Notations and preliminaries

NT	Cross reading
Band $[-A, A]$	Bandwidth (signal)
Cos^2 taper	Apodization (plateau/transition)
Ξ_A	Non-uniform sampling
$\log p/p^{k/2}$	p -adic energy weighting
$C_{\text{fr}}^{(w)} \geq 1 - \Theta_w$	Frame conditioning (AO)
Wave-packet $\hat{\phi}_T$	Matched filter (signal/phys)
Explicit formula	Source-spectrum coupling

3 Storyboard T1–T4

T1: set the arena (PW, explicit, Archimedean).

T2: gap via locality + weights.

T3: packet $\rightarrow$ contradiction if off-line zero.

T4: HP-compatible reading (no postulate).

4 T2 — Weighted frame bound (full)

4.1 Weighted frame bound (T2): full proof

4.1.1 Statement

Let $\phi \in \mathcal{PW}$ be even, $\text{supp } \hat{\phi} \subset [-A, A]$, $\hat{\phi}(0) = 1$.

Let $\Xi_A = \{\xi_{p,k} = k \log p \leq A\}$ and weights $w_{p,k} = \frac{\log p}{p^{k/2}}$.

Define $G_{\alpha\beta}^{(w)} = \sqrt{w_\alpha} \hat{\phi}(\xi_\alpha - \xi_\beta) \sqrt{w_\beta}$.

Theorem 4.1. *There exist $A_0 \geq 1$ and $\Theta_w(A) \in (0, 1)$ for $A \geq A_0$ such that $\lambda_{\min}(G^{(w)}) \geq 1 - \Theta_w(A) =: C_{\text{fr}}^{(w)}(A) > 0$.*

4.1.2 Schur/Gershgorin

Let $R = G^{(w)} - I$. Schur's test gives

$$\|R\|_{2 \rightarrow 2} \leq \left(\sup_{\alpha} \sum_{\beta} |R_{\alpha\beta}| \right)^{1/2} \left(\sup_{\beta} \sum_{\alpha} |R_{\alpha\beta}| \right)^{1/2}.$$

Since R is symmetric,

$$\|R\|_{2 \rightarrow 2} \leq \Theta_w(A) := \sup_{\alpha} \sum_{\beta \neq \alpha} \sqrt{w_\alpha w_\beta} |\hat{\phi}(\xi_\alpha - \xi_\beta)|.$$

Hence $\lambda_{\min}(G^{(w)}) \geq 1 - \Theta_w(A)$.

4.1.3 Locality and summability

Fix $\alpha = (p, k)$. Band-limitation implies $\hat{\phi}(\xi_\alpha - \xi_\beta) = 0$ unless $|k \log p - \ell \log q| \leq A$. Split $\sum_{\beta \neq \alpha} = \sum_{q=p} + \sum_{q \neq p}$.

Same prime $q = p$. Then $|\ell - k| \leq \lfloor A/\log p \rfloor$ and $\sqrt{w_{p^k} w_{p^\ell}} = (\log p)/p^{(k+\ell)/4} \leq (\log p)/p^{k/2} \cdot p^{-(\ell-k)/4}$, whence

$$\sum_{\substack{\ell \neq k \\ |k-\ell| \leq A/\log p}} \sqrt{w_{p^k} w_{p^\ell}} |\hat{\phi}(\xi_{p,k} - \xi_{p,\ell})| \leq \frac{C_1}{p^{k/2-1/4}}.$$

Different primes $q \neq p$ (refined). Locality forces $q^\ell \in [p^k e^{-A}, p^k e^A]$, so $q \leq p^k e^A$ and the number of admissible primes is $\ll p^k e^A / \log(p^k e^A)$ (PNT). For each q , admissible ℓ lie in an interval of length $\ll A / \log q$. Using $\sqrt{w_{p^k} w_{q^\ell}} \leq \sqrt{\log p \log q} p^{-k/4} q^{-\ell/4}$ and $\sum_{q \leq X} \frac{\sqrt{\log q}}{q^{1/4}} = O(X^{3/4})$,

$$\sum_{q \neq p} \sum_{\ell \text{ admissible}} \sqrt{w_{p^k} w_{q^\ell}} |\hat{\phi}(\xi_{p,k} - \xi_{q,\ell})| \leq \frac{C_2 A}{p^{k/4}} (p^k e^A)^{3/8} = C_3 A p^{k/8} e^{3A/8}.$$

Weighted by $w_\alpha = (\log p) / p^{k/2}$ yields $\ll A e^{3A/8} p^{-3k/8}$, geometrically small in k .

Conclusion. Thus

$$\sum_{\beta \neq \alpha} \sqrt{w_\alpha w_\beta} |\hat{\phi}(\xi_\alpha - \xi_\beta)| \leq \frac{C_1}{p^{k/2-1/4}} + \frac{C_3 A e^{3A/8}}{p^{3k/8}},$$

so $\Theta_w(A) < 1$ for $A \geq A_0$ and $\lambda_{\min}(G^{(w)}) \geq 1 - \Theta_w(A) > 0$.

5 T3 — Wave-packet localization (full)

5.1 Wave-packet localization (T3): full proof

Assume Theorem 4.1 and the explicit formula (Appendix). Let $\gamma_0 \in (0, A)$ and suppose a nontrivial zero $\rho_0 = \beta_0 + i\gamma_0$ with $\beta_0 \neq \frac{1}{2}$.

5.1.1 Wave-packet construction

Pick $b \in C_c^\infty([-1, 1])$, even, $\int b = 1$. For $T \geq 1$ set

$$\hat{\phi}_T(\gamma) = \frac{1}{2} \left(b(\sqrt{T}(\gamma - \gamma_0)) + b(\sqrt{T}(\gamma + \gamma_0)) \right).$$

Then $\hat{\phi}_T$ concentrates near $\pm\gamma_0$ with width $\sim T^{-1/2}$; let $\phi_T = \mathcal{F}^{-1} \hat{\phi}_T$, with $\|\phi_T\|_2 = 1$.

5.1.2 Frame capture and discrete-to-continuous limit

By T2, $\sum_{\xi \in \Xi_A} w_{p,k} |\hat{\phi}_T(\xi)|^2 \geq C_{\text{fr}}^{(w)}(A)$. Moreover:

Lemma 5.1 (Frame-to-integral).

$$\sum_{\xi \in \Xi_A} w_{p,k} |\hat{\phi}_T(\xi)|^2 \longrightarrow \int_{-A}^A |\hat{\phi}_T(\gamma)|^2 d\gamma = 1 \quad (T \rightarrow \infty).$$

5.1.3 Archimedean term $\rightarrow 0$ (explicit DCT)

Write $A(\phi) = \int K(t) \hat{\phi}(t) dt$ with $K(t) = 2 \log |t| + O(|t|^{-2})$. Since $\text{supp}(\hat{\phi}_T) \subset \{|t - \gamma_0| \leq c/\sqrt{T}\}$ and $\|\hat{\phi}_T\|_1 = O(1)$, choose

$$h(t) = C (1 + t^2)^{-1} (1 + \log^2(2 + |t|)) \mathbf{1}_{|t - \gamma_0| \leq 2}(t) \in L^1,$$

and note $|K(t) \hat{\phi}_T(t)| \leq h(t)$ for T large; as $\hat{\phi}_T(t) \rightarrow 0$ a.e., DCT gives $A(\phi_T) \rightarrow 0$.

5.1.4 Prime–power side $\rightarrow 0$ (no square)

The explicit formula involves $\sum_{p,k} (\log p)/p^{k/2} \phi_T(\xi_{p,k})$ (not squared). By Fourier inversion and repeated integration by parts, for any $M \in \mathbb{N}$ there exists C_M with

$$|\phi_T(x)| \leq C_M \frac{T^{(M-1)/2}}{(1+|x|)^M}.$$

For fixed A there are finitely many $\xi_{p,k} \in [0, A]$, hence

$$\sum_{p,k} \frac{\log p}{p^{k/2}} |\phi_T(\xi_{p,k})| \leq \max_{\xi \in \Xi_A} |\phi_T(\xi)| \sum_{p,k: \xi_{p,k} \leq A} \frac{\log p}{p^{k/2}} \xrightarrow{T \rightarrow \infty} 0,$$

since $\max_{\xi \in \Xi_A} |\phi_T(\xi)| \rightarrow 0$ (width $T^{-1/2}$, amplitude $\rightarrow 0$ pointwise).

5.1.5 Contradiction

The explicit formula yields

$$\sum_{\rho} \hat{\phi}_T(\gamma_{\rho}) = A(\phi_T) + \sum_{p,k} \frac{\log p}{p^{k/2}} \phi_T(\xi_{p,k}).$$

Letting $T \rightarrow \infty$, LHS $\rightarrow 1$, RHS $\rightarrow 0$; contradiction.

A Guinand–Weil explicit formula

For $\phi \in \mathcal{PW}$ even, smooth, band-limited, contour-shift and residues for the completed ξ -function yield

$$\sum_{\rho} \hat{\phi}(\gamma_{\rho}) = \int K(t) \hat{\phi}(t) \frac{dt}{2\pi} + \sum_{p,k \geq 1} \frac{\log p}{p^{k/2}} \phi(k \log p),$$

with $K(t) = \psi(\frac{1}{4} + \frac{it}{2}) + \psi(\frac{1}{4} - \frac{it}{2}) - 2\psi(\frac{1}{4})$ and $\psi = \Gamma'/\Gamma$.

B Archimedean control (IBP)

Using $\psi(\frac{1}{4} \pm \frac{it}{2}) = \log |t| + O(|t|^{-2})$ uniformly and a $\cos^2$ taper with plateau+transition, two integrations by parts on the boundary layer give

$$|A(\phi)| \leq \frac{C}{A^2} \|\hat{\phi}\|_1.$$

C Absorption lemma

If $a \leq \eta a + b$ with $\eta \in (0, 1)$, then $a \leq b/(1 - \eta)$.

D FAQ (by specialty)

NT: why $p^{-k/2}$? — damp prime powers.

AO: why Schur? — direct control of $\|R\|$.

Signal: why $\cos^2$? — bounded slope, simple calculus.

Phys: HP? — reading only.

Repro: numbers? — certificates, not part of the proof.

About the Author

InterIA Mathematical Collective — Couret – Unification Series, Vol. 1. This repository inaugurates a series of open, reproducible mathematical artefacts led by Alexandre Couret and collaborators, formalizing analytic proof frameworks around classical number-theoretic problems.

The present volume packages the T1–T4 analytic chain (explicit formula, weighted frame bound, wave-packet localization, cross-positivity) as a journal-ready LaTeX project with complete proofs and build recipes. It is a *framework proof*, not a resolution of RH.

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